

Homework 24: Elementary Matrices & Inverses Using Elementary Matrices

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BCS A, E, F, G

Definition 1. A square matrix $A \in M_n$ is called an *elementary matrix* if it can be obtained from the identity matrix via a single elementary row operation.

Remark 1. Note that multiplying on the left by an elementary matrix always gives the same result as applying the corresponding elementary row operation.

Problem 1. A Matrix Decomposition Based on the RREF.

Let

$$A = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$$

Find a 3×3 matrix E such that $EA = R$, where R is the reduced row echelon form of A . Note that the matrix E is the product of all elementary matrices $E_i \forall i$, that are required to produce reduced row echelon form of A .

Theorem 1. If $A, R \in M_{m,n}$ and $E \in M_m$ are matrices such that the block matrix $[A|I]$ can be row-reduced to $[R|E]$, then $R = EA$.

Let us use this theorem to revisit Problem 1.

Problem 2. An Alternative Method to Find E .

Let

$$A = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$$

Compute the reduced row echelon form of the 1×2 block matrix $[A|I]$.

The Inverse of a Matrix

One of the nice features of the elementary matrices is that they are invertible: multiplication by them can be undone by multiplying by some other matrix. For example, if

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix},$$

then it is straightforward to check that $E_1 E_2 = I$ and $E_2 E_1 = I$. We thus say that E_1 and E_2 are inverses of each other, which makes sense in this case because E_1 is the elementary matrix corresponding to the elementary row operation $R_3 - 3R_1$, and E_2 is the elementary matrix corresponding to the elementary row operation $R_3 + 3R_1$, and performing these row operations one after another has the same effect as not doing anything at all.

Definition 2. Invertible Matrices.

A square matrix $A \in M_n$ is called *invertible* if there exists a matrix, which we denote by A^{-1} and call the *inverse* of A , such that

$$AA^{-1} = A^{-1}A = I.$$

Problem 3. Inverses of Elementary Matrices.

Find the inverse of each of the following elementary matrices.

a. $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$.

b. $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.

c. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

d. $\begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

Theorem 2. Suppose $A, B \in M_n$ are invertible. Then AB is also invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Problem 4. Finding an Inverse of a Product.

Use “Theorem 2” and “inverses from Problem 3” to find the inverse of the matrix

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}.$$

Theorem 3. Finding Matrix Inverses.

Suppose $A \in M_n$. Then A is invertible if and only if there exists a matrix $E \in M_n$ such that the RREF of the block matrix $[A|I]$ is $[I|E]$. Furthermore, if A is invertible then it is necessarily the case that $A^{-1} = E$.

Problem 5. For each of the following matrices, either compute their inverse or show that none exists.

a. $\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}$.

b. $\begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix}$.

c. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

d. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$.

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Good Luck

Appendix. One Sided Inverses.

Theorem A. Suppose $A \in M_n$. If there exists a matrix $B \in M_n$ such that $AB = I$ or $BA = I$ then A is invertible and $A^{-1} = B$.

Remark I. One-Sided and Non-Square Inverses.

When we defined inverses in Definition 2, we required that the matrix A was square. Some of the ideas from this section can be extended to the non-square case, but the details are more delicate. For example, consider the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

This matrix is not square, and thus not invertible, but it does have a “**left**” inverse—there is a matrix B such that $BA = I$. For example,

$$BA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

No Right Inverse For A. However, it does not have a “**right**” inverse: there is no matrix C such that $AC = I$, since the only way that AC could be square is if C is 2×3 , in which case the third row of AC must be the zero vector.

Remark II. In fact, A has infinitely many left inverses—we can replace the entries of the third column of B by whatever we like.

Remark III. The fact that A has a left inverse but not a right inverse does not violate “*Theorem A*”, since that theorem only applies to square matrices.